

BCNmodular

MAESTRO & SomeStructure

Probabilistic Voice Router & Arrangement Structure Generator

User Manual v1 | Plugin version 2.2.0

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1. Introduction

1.1 The BCNmodular system

BCNmodular is a plugin for VCV Rack 2 that brings musical intelligence to modular synthesis. It consists of two complementary modules designed to work together:

- **Maestro** — the probabilistic voice router. It decides which of your voices should be active at each evaluation point, based on weighted probability, density control, and musical timing.
- **SomeStructure** — the arrangement structure generator. It defines the large-scale form of your track: which sections play in which order, how dense each section should be, and which roles each voice should take.

Together, they create a system where **SomeStructure provides the structure** and **Maestro provides the life**. The result is generative music with real musical form — sections that feel intentional, with natural variation within each section.

Both modules can also be used independently. Maestro works as a powerful standalone probabilistic router, and SomeStructure can send its CV outputs to any module.

1.2 System requirements

- VCV Rack 2.x (Free or Pro)
- Compatible with all operating systems supported by VCV Rack

2. Installation

2.1 From the VCV Library (recommended)

- Open VCV Rack 2
- Click Library in the top menu
- Search for BCNmodular
- Click Subscribe and then Update plugins
- Restart VCV Rack

2.2 From GitHub (manual)

Download the latest .vcvplugin file for your platform from:

`github.com/santifort-commits/BCNmodular/releases`

Place the file in your VCV Rack plugins folder:

- Windows: Documents\Rack2\plugins
- Mac: ~/Documents/Rack2/plugins
- Linux: ~/Documents/Rack2/plugins

MAESTRO

Probabilistic Voice Router, Arranger, Mixer, Sequencer & Trigger Generator



Maestro panel — v2.2.0

3. MAESTRO

Maestro is an arrangement tool for VCV Rack 2 that brings controlled randomness to your modular patches. Instead of building fully deterministic sequences, Maestro acts as an intelligent conductor, deciding which voices play at each moment based on weighted probability, density control, and musical timing.

In modular synthesis, achieving musical structure while preserving randomness typically requires combining many modules. Maestro consolidates this into a single module: connect your voices, set their relative probabilities, and let Maestro decide who plays at each evaluation point.

What Maestro is NOT

- It is not a sequencer in the traditional sense — it does not play notes
- It is not a mixer — it does not control volume levels
- It is not a random module — its randomness is musically controlled

3.1 Quick Start

Your first Maestro patch

Step 1: Add Maestro to your rack — right-click on an empty rack area and search for Maestro.

Step 2: Connect a clock — connect your master clock output to the CLOCK input (top left of Maestro).

Step 3: Connect your voices — connect the gate or audio output of your first voice to CH1 INPUT. Connect CH1 OUTPUT back to your voice's destination. Repeat for up to 6 voices.

Step 4: Set basic parameters:

- ACTIVE TRACKS: set to the number of voices you have connected (e.g. 3)
- TRACK DENS: set to how many voices you want active at once (e.g. 2)

- **LENGTH:** set to 1 or 2 bars for quick evaluation

Step 5: Start your clock and listen. Maestro will now periodically decide which voices are active. Watch the LEDs to see which channels are open (green) or closed (red).

TIP

Start with Gate mode (G) for all channels. Once you understand the basic behaviour, experiment with Fade mode (F) for audio channels and Trigger mode (T) for triggering events.

3.2 Panel Overview

Maestro is organized in three zones:

- Row 1 (Global controls): Clock, density, and output settings
- Row 2 (Timing controls): Length, skip, and fade parameters
- Channel rows (x6): Individual channel controls

Global controls (Row 1)

Control	Type	Description
CLOCK	Input	Clock/trigger input. Drives the bar counter. Connect to your master clock.
ACTIVE TRACKS	Knob	Number of channels participating in evaluation (1 to 6).
TRACK DENS	Knob	Base number of active voices. This is the target number of channels Maestro will try to keep active.
TRK DENS XFADE	Knob	Crossfade between Track Density knob and DENS CV. 0 = knob only, 1 = CV only.
DENS CV	Input	CV input for density modulation (0–10V). Balance with TRK DENS XFADE.
DETERM/RANDOM	Knob	Randomness. Fully left (D) = deterministic. Fully right (R) = maximum gaussian distribution.
OUT CH. AMOUNT	Output	CV output proportional to number of active voices (0–10V).

Timing controls (Row 2)

Control	Type	Description
RESET	Input	Resets the bar counter (default) or forces an immediate evaluation. Configurable in context menu.
LENGTH	Knob	Evaluation period in bars: 1, 2, 4, 8 or 16 bars.
LEN CV	Input	CV input for length. When connected, overrides the LENGTH knob.
SKIP PROB	Knob	Probability of skipping an evaluation (0 = never skip, 1 = always skip).
SKIP CV	Input	CV input for skip probability.
FADE IN	Knob	Fade-in time for channels in Fade mode (0 to 10 seconds).
FADE OUT	Knob	Fade-out time for channels in Fade mode (0 to 10 seconds).

Per-channel controls (x6)

Control	Type	Description
LABEL	Text	Editable 4-character label. Double-click to edit. Saved with the patch.
INPUT	Input	Signal input: gate, clock, audio or trigger.
PROB	Knob	Base probability weight for this channel (0 to 1).
PROB XFADE	Knob	Crossfade between Prob knob and PROB CV. 0 = knob only, 1 = CV only. Labels Knob/CV indicate blend direction.

PROB CV	Input	CV modulation for the channel probability (0–10V).
T/G/F	Switch	Channel mode: T = Trigger, G = Gate, F = Fade.
LED	Light	RGB status indicator. Colours depend on channel mode.
OUT	Output	Signal output. Passed or blocked according to channel state and mode.

3.3 Channel Modes (T/G/F)

Each channel has an independent mode switch with three positions.

Gate mode (G)

Gate mode provides instant on/off switching. When a channel is selected by Maestro, the input signal passes through immediately. When deselected, the signal is blocked immediately.

Best for: clock and gate signals going to sequencers, controlling external switches or VCAs, any signal where instant response is desired.

LED behaviour: Green = channel active. Red = channel inactive.

Fade mode (F)

Fade mode applies a smooth amplitude envelope when opening or closing a channel. The fade-in and fade-out times are set globally by the FADE IN and FADE OUT knobs.

Best for: audio signals (avoids clicks and pops), preserving natural reverb tails, smooth transitions between voices.

LED behaviour: Green = fully open. Yellow = fade in progress. Red = fully closed.

TIP

Use long FADE OUT times (3–10 seconds) when your voices have natural reverb or delay tails. This prevents abrupt cuts that would sound unnatural.

Trigger mode (T)

Trigger mode sends a short 10V pulse each time the channel is selected by an evaluation. The signal is only 'on' for a very brief moment (1–10ms, configurable).

Best for: firing burst generators and envelope generators, triggering samplers and drum modules, creating probabilistic percussion patterns.

LED behaviour: Blue (constant) = trigger mode, waiting. White flash = trigger fired.

Important: All selected channels fire simultaneously. The trigger duration can be adjusted in the context menu (1ms, 2ms, 5ms default, 10ms).

TIP

If a module does not respond to Maestro triggers, try increasing the trigger duration to 10ms in the context menu. Some modules need longer pulses.

3.4 Prob CV Crossfade

The PROB XFADE knob controls the balance between the PROB knob and the PROB CV input:

- XFADE = 0 → the knob value is used entirely (CV has no effect)
- XFADE = 0.5 → the result is a blend of knob and CV
- XFADE = 1 → the CV value is used entirely (knob has no effect)

This also applies to TRACK DENSITY via the TRK DENS XFADE knob.

The Prob CV crossfade mode (context menu) controls how the crossfade responds:

Mode	Formula	Description
Normal (Linear)	w^1	Standard linear response. Best for manual use without SomeStructure.
Biased (Quadratic)	w^2	Emphasises differences between high and low weights. Good balance.
Hard (Cubic)	w^3	Highly deterministic. Recommended when using SomeStructure to control Maestro.

TIP When using SomeStructure, set Prob CV crossfade mode to Hard (Cubic) and PROB XFADE to 1. This ensures SomeStructure's voice role weights are followed as closely as possible.

3.5 Context Menu

Right-click on Maestro to access advanced settings. The context menu is organised in three groups:

Group 1: How it evaluates

Option	Values	Description
Prob CV crossfade mode	Normal / Biased / Hard	Controls how strongly Prob CV weights influence voice selection.
Beats per bar	2–8 (default: 4)	Sets the time signature. LENGTH is counted in bars, each bar has this many beats.
Min active voices	0–6 (default: 0)	Minimum number of channels that must be active at all times.

Group 2: When it evaluates

Option	Values	Description
Skip CV mode	Probabilistic / Binary	Probabilistic: SKIP CV sets probability. Binary: below 1V = never skip, above 1V = always skip.
Skip mode	Global / Per channel	Global: entire evaluation skipped or not. Per channel: each channel decides independently.
Reset input mode	Reset bars / Force evaluate	Reset bars: resets bar counter. Force evaluate: triggers immediate evaluation without resetting.

Group 3: How it sounds

Option	Values	Description
Default input voltage	1V (default) / 10V	Voltage when no input is connected. Use 1V for gates/triggers, 10V for CV control.
Active output CV mode	Proportional / Absolute	Proportional: scales to ACTIVE TRACKS. Absolute: always scales to 6 voices.
Trigger duration	1ms, 2ms, 5ms, 10ms	Duration of the trigger pulse. Increase if modules do not respond reliably.

3.6 Typical Use Cases

Arrangement tool

Connect the output of multiple sequencers or voice modules to Maestro inputs. Set LENGTH to 2 or 4 bars. Maestro will periodically decide which voices are active, creating evolving arrangements that never repeat exactly.

Performance tool

Set Reset input mode to Force evaluate (context menu), and connect a momentary button or footswitch to RESET. Press RESET to trigger a new arrangement at any moment during performance.

Probabilistic trigger sequencer

Set all channels to Trigger mode and leave inputs unconnected. Connect outputs to burst generators, envelope generators, or drum modules. Each evaluation, Maestro decides which instruments fire their trigger.

Smooth audio mixer

Connect audio outputs from different voices to Maestro inputs. Use Fade mode with long fade times for smooth crossfades between sounds.

Self-regulating patch

Connect the OUT CH. AMOUNT output back to the DENS CV input. This creates a feedback loop where the current density influences future density decisions.

3.7 Tips & Tricks

TIP Use channel labels (double-click) to name your voices: KICK, BASS, PAD, LEAD, etc. This makes complex patches much easier to understand at a glance.

TIP Set DETERM/RANDOM fully left for exact voice counts. Turn it right gradually to introduce variation in the number of active voices.

TIP In 3/4 or 6/8 time, set Beats per bar to 3 or 6 in the context menu to ensure Maestro evaluates on musically correct boundaries.

TIP Use MIN ACTIVE VOICES in the context menu when you always need at least one voice playing (e.g. bass or kick drum should never stop).

TIP Longer FADE OUT times (5–10 seconds) work beautifully with reverb-heavy pads — the reverb tail fades naturally as the next voice comes in.

TIP In Force evaluate mode, the RESET input becomes a performance controller. Map it to a MIDI button or footswitch to manually trigger arrangement changes.

TIP Mix channel modes: use Gate for clock signals, Fade for audio, and Trigger for percussion.

3.8 Technical Reference

Signal levels

Signal	Range	Notes
CLOCK input	> 1V = high	Rising edge triggers beat count
RESET input	> 1V = trigger	Rising edge only
DENS CV input	0 to 10V	Balanced by TRK DENS XFADE
LEN CV input	0 to 10V	Overrides LENGTH knob when connected
SKIP CV input	0–10V (prob) / 0–10V (binary)	See Skip CV mode
PROB CV inputs	0 to 10V	Balanced by PROB XFADE per channel
OUT CH. AMOUNT	0 to 10V	Proportional or absolute (context menu)
Trigger output (T mode)	10V pulse	Duration: 1–10ms (configurable)

Gate output (G mode)	Input voltage	Passes or blocks input signal
Fade output (F mode)	0 to input voltage	Smooth amplitude envelope

The evaluation algorithm

When an evaluation is triggered, Maestro performs the following steps:

- 1. Calculate target voices:** TRACK DENS crossfaded with DENS CV (per TRK DENS XFADE), clamped to [Min active voices, ACTIVE TRACKS].
- 2. Apply randomness:** if DETERM/RANDOM > 0, the target is varied using a gaussian distribution.
- 3. Weighted random selection:** channels are selected using weighted random sampling. Each channel's weight is its PROB knob crossfaded with PROB CV, then raised to the power defined by the Prob CV crossfade mode (1, 2, or 3).
- 4. Apply skip probability:** in Global mode, the entire evaluation may be skipped. In Per channel mode, each channel independently may retain its previous state.
- 5. Timing:** all evaluations are delayed by one audio sample to allow CV signals from external modules to fully stabilise.

Saving and loading

The following settings are saved with your VCV Rack patch:

- All knob positions and mode switches (T/G/F)
- All channel labels
- All context menu settings: Prob CV crossfade mode, beats per bar, min active voices, default input voltage, active output CV mode, trigger duration, skip CV mode, skip mode, reset input mode

SomeStructure

Arrangement Structure Generator for Maestro



SomeStructure panel — v2.2.0

4. SOMESTRUCTURE

SomeStructure is a companion module for Maestro. While Maestro handles probabilistic voice routing at the bar level, SomeStructure provides the higher-level musical structure: it defines which sections a track passes through (Intro, Build, Drop, Outro...) and how the voice density and roles should evolve across those sections.

SomeStructure provides the structure. Maestro provides the life.

4.1 Quick Start

Step 1: Add SomeStructure and Maestro to your rack. Connect both to the same master clock.

Step 2: Connect SomeStructure outputs to Maestro inputs:

- DENSITY CV → Maestro DENS CV
- CH.1–CH.6 → Maestro PROB CV CH1–CH6
- SECTION TRIGGER → Maestro RESET

Step 3: In Maestro, set:

- TRK DENS XFADE = 1 (CV controls density)
- PROB XFADE = 1 for all channels (CV controls probabilities)
- Reset input mode = Force evaluate (context menu)
- Prob CV crossfade mode = Hard (Cubic) (context menu)

Step 4: Select a style in SomeStructure (e.g. Electronic → Drum & Bass) and start your clock.

TIP

Start with Duration = 1 bar for all parts and LENGTH = 1 bar in Maestro. This gives the tightest synchronisation between structure and voice selection.

4.2 Panel Overview

Left column — Global controls

Control	Type	Description
CLOCK	Input	Clock input. Should be the same clock as Maestro.
RESET	Input	Resets to section 1 and restarts the arrangement.
ADV. MODE	Switch	MAN. = Manual (only Jump buttons advance). AUTO = Automatic progression.
STYLE	Switch	Style category: CUSTOM / ELEC. (Electronic) / ORG. (Organic).
GENRE display	Display	Shows the active genre name.
GEN SEL	Knob	Selects the genre within the active category.
PART BLUR	Knob	Adds looseness to section durations. Small values keep structure tight.
BLUR ATT.	Knob	Attenuverter for the Blur CV input.
BLUR CV	Input	CV input for blur amount.

Part Configuration Area

Control	Description
ACT (LED)	Green LED indicates the currently active section.
PART (button)	Section name label. In Custom style, right-click to configure name, density and voice weights.
DURATION (knob)	Section duration: 0=Skip, 1=1 bar, 2=2 bars, 4=4 bars, 8=8 bars. Duration 0 (Skip) bypasses the section entirely.
JUMP (button)	Jump to this section. Waits for the next bar boundary (intentional — ensures musical timing). MIDI-mappable.

Recommended Voice Display

The top panel area shows the recommended instrument for each density channel (CH1–CH6) based on the active style. This updates automatically when you change style or genre.

Right column — Outputs

Output	Description
BUS OUT	16-channel polyphonic BCNmodular Bus. Carries all structural parameters.
DENSITY CV	Density target for the current section (0–10V = 0–6 voices).
SECTION CV	Current section number as CV (0–10V = sections 1–8).
SECTION TRIGGER	Trigger pulse at every section change.
PHRASE TRIGGER	Trigger pulse every N bars (configurable in context menu, default 1 bar).
EOC	End of cycle trigger — fires when the arrangement returns to section 1.
CH.1–CH.6	Individual role weight CV per channel (0–10V = 0.0–1.0 weight).

4.3 Styles & Genres

Category	Genres
Electronic	Techno, Drum & Bass, EDM, IDM/Experimental, Synthwave, Dubstep
Organic	Prog/Symphonic Rock, Jazz, Ambient, Minimal, Dub
Custom	Custom A, Custom B, Custom C (user-editable, saved per patch)

Each genre defines 8 sections with a name, a density target, a role weight for each channel, and recommended voice names.

4.4 Custom Styles

Custom A, B, and C are fully user-editable styles. In Custom mode, right-click on any PART button to access its configuration:

Menu item	Description
Name	Section type name (Intro, Vers, PreCh, Chorus, Bridge, Break, BrkDn, Outro, Custom).
Density	Target number of active voices (1–6).
Voice weights → CH1–CH6	Role weight per channel: 0.0 (never), 0.1 (rarely), 0.4 (occasionally), 0.7 (often), 1.0 (always).
Recommended voice names	Label each channel with an instrument name.

Custom configurations are saved per patch and per preset. Custom A, B, and C are completely independent — you can configure each one differently and switch between them without losing your settings.

4.5 Context Menu

Option	Values	Description
Phrase trigger every	1, 2, 3, 4, 6, 8 bars (default: 1)	How often the PHRASE TRIGGER output fires. Useful for syncing external modules to musical phrase boundaries.
Beats per bar	2–8 (default: 4)	Sets the time signature. Must match Maestro's beats per bar setting.

4.6 BCNmodular Bus

The BUS OUT output carries a 16-channel polyphonic signal with all structural parameters:

Channel	Content
1	Density target (0–10V = 0–6 voices)
2	Randomness blend
3	Length override
4	Section blur
5	Section CV (0–10V = sections 1–8)
6	Section trigger
7	Phrase trigger
8	End of cycle
9	Tension CV
10	Protocol version (1.0)
11–16	Role weight CH1–CH6 (0–10V)

4.7 Tips & Tricks

TIP	Use Duration = 0 (Skip) to create songs with fewer than 8 active sections. Place Skip on the last unused sections.
TIP	Jump buttons can be MIDI-mapped. Right-click on any Jump button and select "Map to MIDI". This allows live performance control with MIDI pads — one pad per section.

TIP

The Jump button waits for the next bar boundary before advancing. This is intentional — section changes always happen on a musically correct beat, even if you press Jump mid-bar.

TIP

Connect PHRASE TRIGGER to reset external sequencers or arpeggiators every N bars, keeping them in sync with the SomeStructure arrangement.

TIP

Connect EOC to the RESET input of SomeStructure itself for infinite looping arrangements.

TIP

In Manual mode (ADV. MODE = MAN.), the arrangement only advances when you press Jump buttons. Ideal for live performance where you want full control over the arrangement flow.

TIP

Beats per bar in SomeStructure must match Maestro's beats per bar setting. Both default to 4.

5. Using MAESTRO + SOMESTRUCTURE together

The combination of both modules creates a complete generative arrangement system. SomeStructure defines what the music should sound like at each stage; Maestro decides exactly which voices play at each moment within that stage.

5.1 Connection guide

From (SomeStructure)	To (Maestro)	Notes
DENSITY CV	DENS CV	Set TRK DENS XFADE = 1
CH.1	PROB CV CH1	Set PROB XFADE = 1 on CH1
CH.2	PROB CV CH2	Set PROB XFADE = 1 on CH2
CH.3	PROB CV CH3	Set PROB XFADE = 1 on CH3
CH.4	PROB CV CH4	Set PROB XFADE = 1 on CH4
CH.5	PROB CV CH5	Set PROB XFADE = 1 on CH5
CH.6	PROB CV CH6	Set PROB XFADE = 1 on CH6
SECTION TRIGGER	RESET	Set Reset mode = Force evaluate

5.2 Recommended Maestro settings for SS-driven mode

Parameter	Value	Reason
TRK DENS XFADE	1	SomeStructure controls density entirely
PROB XFADE (all channels)	1	SomeStructure controls voice roles entirely
TRACK DENS knob	any	Overridden by DENSITY CV
PROB knobs (all channels)	0.5	Neutral — overridden by PROB CV
DETERM/RANDOM	fully left	Deterministic voice counts
Prob CV crossfade mode	Hard (Cubic)	Maximum responsiveness to role weights
Reset input mode	Force evaluate	Re-evaluate immediately on section change
Beats per bar	match SS	Must match SomeStructure beats per bar

5.3 Understanding the system

SomeStructure is deterministic — each section always sends the same CV values for a given style. If section 3 is a "Drop" with density 5 and CH1=1.0, CH2=1.0, CH3=0.7, CH5=0.7, those values are always the same every time.

Maestro is probabilistic — it uses those CVs as weighted probabilities. So within a Drop section with those role weights, Maestro might choose CH1+CH2+CH3+CH4+CH5 one time, and CH1+CH2+CH3+CH5+CH6 the next. The distribution is biased towards higher-weight channels, but there is natural variation.

The result — musical structure with organic variation. The arrangement has clear sections that feel intentional, but no two playthroughs are identical.

TIP

Use Prob CV crossfade mode = Hard (Cubic) for maximum determinism with SomeStructure. In Hard mode, a channel with weight 0.1 is 1000 times less likely to be selected than a channel with weight 1.0.

TIP

You can override SomeStructure's influence at any time by reducing the PROB XFADE or TRK DENS XFADE knobs. This lets you blend between SomeStructure-driven and manually-driven arrangements in real time.

TIP

Connect SomeStructure SECTION TRIGGER → Maestro RESET (Force evaluate) to ensure Maestro always re-evaluates immediately when a new section starts, using the correct CVs for that section.

6. Changelog

Version 2.2.0

- **New module: SomeStructure** — arrangement structure generator
- 8 configurable sections with name, duration (Skip/1/2/4/8 bars) and Jump button
- 11 built-in styles across Electronic and Organic categories
- 3 user-editable Custom styles (A, B, C) saved per patch and preset
- Individual CV outputs: Density, Section CV, Section Trigger, Phrase Trigger, EOC, CH1–CH6
- BCNmodular Bus — 16-channel polyphonic output with all structural parameters
- Jump buttons with MIDI mapping support
- Phrase trigger configurable: 1, 2, 3, 4, 6, or 8 bars
- Duration 0 = Skip — bypasses section entirely for songs with fewer than 8 parts
- **Maestro improvements:**
- Prob CV and Track Density CV now use crossfade logic (Knob ↔ CV) instead of attenuverters
- Prob CV crossfade mode: Normal (Linear), Biased (Quadratic), Hard (Cubic)
- Evaluations delayed by 1 audio sample for full CV stabilisation
- Context menu reorganised into 3 logical groups: How it evaluates / When it evaluates / How it sounds
- Panel labels updated: TRK DENS XFADE, PROB XFADE, Knob/CV indicators

Version 2.1.1

- Fixed typo "Lenght" → "Length" on panel
- Updated author contact email

Version 2.1.0

- Added Trigger mode (T) per channel: sends a 10V pulse when selected
- RGB LEDs per channel and mode
- Configurable trigger duration via context menu
- Skip mode: Global or Per channel
- Skip CV mode: Probabilistic or Binary
- Reset input mode: Reset bars or Force evaluate
- Fixed: simultaneous triggers now fire correctly on all selected channels

Version 2.0.0

- Initial public release

7. License & Contact

BCNmodular is released under the GPL-3.0-or-later license. The source code is available at:

`github.com/santifort-commits/BCNmodular`

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Bug reports and feature requests: please open an issue on GitHub.